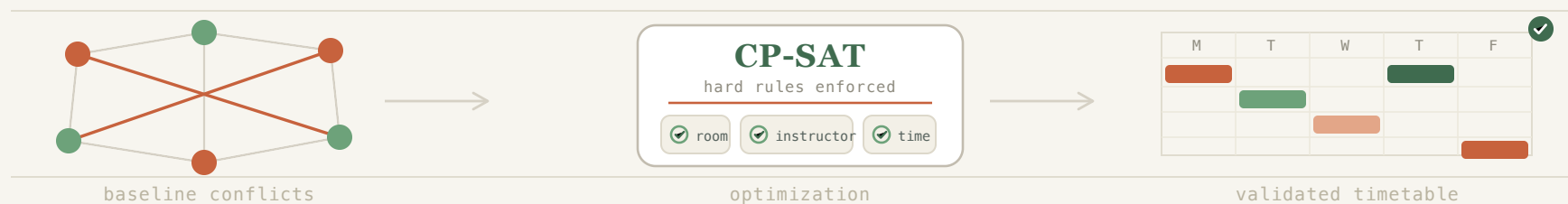


Every hard conflict — **eliminated.** Twice.

A CP-SAT scheduling engine rebuilt from real TED University data — tested end-to-end across two live academic terms.

● — ● BASELINE → OPTIMIZED, BOTH TERMS



FALL 2025



854 → 0

hard conflicts

SPRING 2026



808 → 0

hard conflicts

↘ **−75%** within-day idle time per cohort

↘ **−29%** capacity waste per room-hour

How **KAIROS** works.

From raw registrar data to a verified, conflict-free timetable.

1

Generate legal candidates

Hard rules are pruned up front: capacity, room type, time window, blackout, and instructor availability. Illegal placements are never even generated.



2

Construct a feasible schedule

Each block is greedily placed into its lowest-cost feasible slot.



3

Repair with CP-SAT

Unplaced blocks and their competitors are re-solved in bounded neighbourhoods with CP-SAT, warm-started, until every block is placed.



4

Polish with Great Deluge

Improves idle gaps, instructor weeks and room stability — without ever relaxing a hard constraint.



5

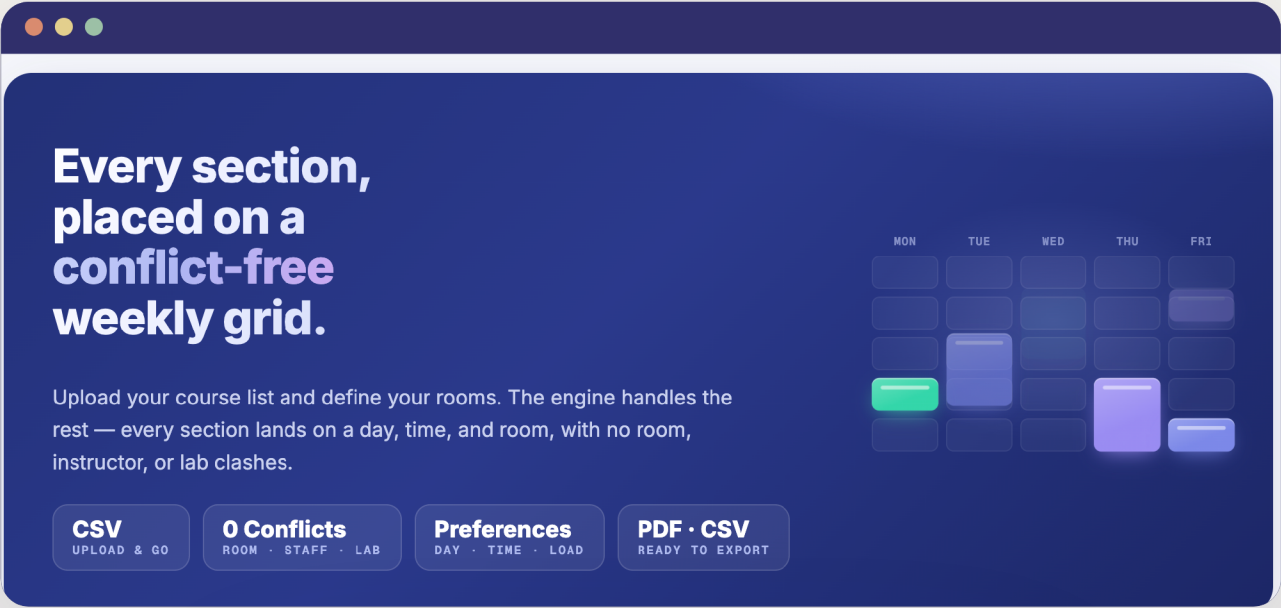
Validate independently

A separate validator re-checks every assignment, decoupled from the solver, so an encoding bug can't pass silently.

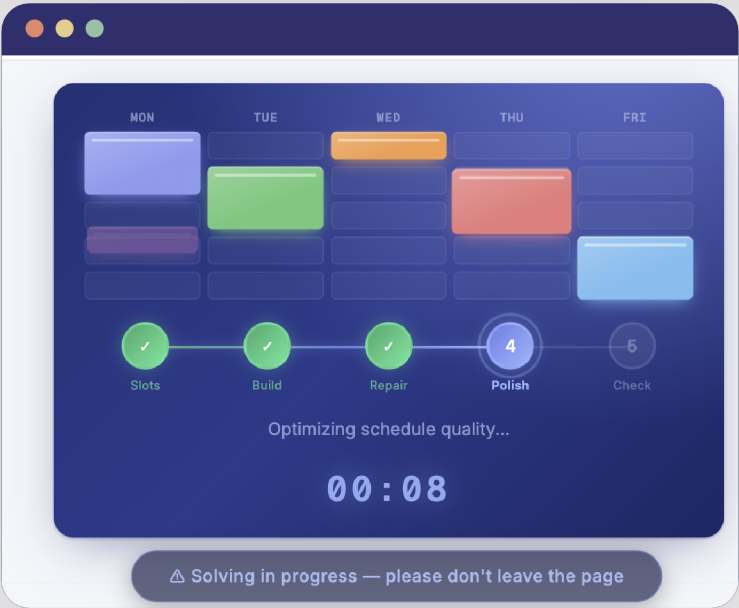


See every placement. Verify every constraint.

A bilingual TR / EN web app — no model expertise required.



 Build a conflict-free schedule



 Repair and optimize every constraint

Feasibility restored. Quality measured.

FEASIBILITY

Every hard-rule category — time-window, instructor, classroom, section, seminar rule, room-type — falls from hundreds of conflicts per term to **zero**, in both terms.



QUALITY KPIS

Across both terms, **idle time and cohort conflicts fall sharply**, while occupancy and evening-course load stay essentially flat.

Instructor conflicts Fall 320 → 0 Spring 314 → 0	Classroom conflicts Fall 339 → 0 Spring 298 → 0	Thursday 14–16h rule Fall 90 → 0 Spring 120 → 0	Potential cohort conflicts Fall 951 → 739 Spring 1040 → 770
Avg. within-day idle time Fall 0.73h → 0.18h Spring 0.79h → 0.18h	Cohort timetable compactness Fall 91% → 98% Spring 90% → 98%	Avg. classroom occupancy Fall 52% → 54% Spring 54% → 54%	Evening-course concentration Fall 14% → 17% Spring 16% → 17%

Not just a solver — a system you can use.

Built end-to-end and running live — inspect the code or try it yourself.

- ✓ All hard conflicts → **zero**, both terms.
- ↘ Idle **-75%** · waste **-29%** per room-hour.
- 🌐 Bilingual **TR / EN** app · runs on user CSVs.

BUILT ON



Python
Core language



Streamlit
TR / EN web app



OR-Tools
Google solver suite



CP-SAT
Exact constraints



Great Deluge
Soft-objective polish



Docker
Cloud Run deploy

🌐 kairos.huguryildiz.com

Try the live app →

🐙 github.com/huguryildiz/KAIROS

Read the code →

👤 huguryildiz.com

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